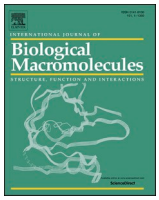
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Valorization of rapeseed straw through the enhancement of cellulose accessibility, lignin removal and xylan elimination using an *n*-alkyltrimethylammonium bromide-based deep eutectic solvent

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ABSTRACT

n-Alkyltrimethylammonium bromide (C_nTAB)-based deep eutectic solvent (DESs) has potential in the efficient delignification and utilization of carbohydrates in biomass. In this research, DESs containing Brønsted acid and Lewis acid were prepared with C_nTAB (alkyl-chain length 12–18), organic acids and metal chlorides, and the optimal treatment conditions were acquired by pretreatment optimization. Through the pretreatment with TTAB/LCA/Fe^{3+} (1:4:0.0111, mol:mol:mol) (162.5 °C, 61.7 min), lignin (89.2 %) and xylan (77.9 %) were effectively eliminated, and the hydrophobicity of rapeseed straw substantially declined from 4.62 to 2.09 m^2/g, acquiring the highest enzymatic saccharification efficiency of 92.5 %. The relationship of DES properties and enzymatic saccharification efficiency was explored. Additionally, hemicellulose in rapeseed straws could be efficiently transformed into furfural (3.75 g/L) and xylo-oligosaccharides (3.64 g/L). To clarify the structural and property changes brought by pretreatment, rapeseed straws were testified by FT-IR, SEM and CLSM and deeply discussed. The interaction between lignocellulose and TTAB/LCA/Fe^{3+} was elucidated by molecular dynamics simulations and quantum chemical calculations, explaining the effectual treatment performance and hemicellulose upgrading at the molecular level. Eventually, a potential pretreatment mechanism of TTAB/LCA/Fe^{3+} was proposed. This established TTAB/LCA/Fe^{3+} treatment holds great promise for valorization of biomass into biofuels and biobased chemicals.

1. Introduction

In the pursuit of sustainable energy, human society is actively seeking alternatives to fossil fuel consumption [1]. One promising approach is the utilization of lignocellulosic biomass, which is composed of carbohydrates (hemicellulose plus cellulose) and lignin, making it suitable for manufacturing biofuels and chemicals [2,3]. Enzymatic digestion is extensively regarded as the most promising method for utilizing carbohydrates [1]. These lignocellulose-derived reducing sugars can be further upgraded to high value-added chemicals or biofuels through fermentation or biocatalysis [4,5]. And these upgraded chemicals include furfural, xylo-oligosaccharides, ethanol, etc., which play an important role in the fields of resin/pharmaceutical intermediates synthesis, health products production and energy [6]. However, the existence of lignin in lignocellulose poses challenges as they form barriers around carbohydrates, blocking the enzymatic digestion process [7]. Thus, developing an effectual pretreatment strategy for breaking down the complex structure of lignocellulosic biomass is crucial. Deep eutectic solvents (DESs) have emerged as effective solvents for lignocellulose treatment due to their excellent performance in enhancing enzymatic digestion [8,9]. They are a kind of readily synthesizable solvent with low vapor pressure, which can effectively separate hemicellulose and lignin from biomass. Compared with traditional hydrothermal pretreatment, it avoids the significant investment in high-pressure reactors and is less corrosive to equipment, reducing subsequent maintenance costs [10]. Typically, choline chloride (ChCl) and betaine (BA) are used as hydrogen bond acceptors (HBAs) in DESs [11]. Besides conventional HBAs, alternative HBAs with additional functionalities for pretreatment hold significant potential to further advance pretreatment technology. The cationic surfactant (e.g., *n*-

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alkyltrimethylammonium bromide, C_nTAB) is one such alternative for HBA of DESs [12]. As an additive with good safety and controllable costs, C_nTAB is widely used in the manufacturing of rubber and daily chemical products. Currently, the acidic CTAB:LA pretreatment strategy developed based on CTAB surfactant has shown a remarkable increase in enzyme digestibility by 72 % and an improvement in cellulose accessibility from 297 mg/g to 473 mg/g [12]. Pretreatment based on CTAB-based DES has been testified to be highly beneficial as it aids in delignification and prevents the redeposition of pseudo-lignin, thereby improving cellulose accessibility and accelerating cellulose hydrolysis. In addition, the metal salt, as a kind of inexpensive and efficient Lewis acid catalysts, has attracted special attention due to its excellent pretreatment performance [13]. As a Lewis acid catalyst, metal salts can not only catalyze the hydrolysis of β-1,4-glucoside, and β-O-4 bonds in hemicellulose and lignin, but also coordinate with hydroxyl and phenolic groups in lignin, change their structure and chemical properties, and improve the efficiency of lignin removal. It has been widely reported to accelerate the hydrolysis of lignocellulosic biomass, particularly hemicellulose, and to significantly improve the efficiency of biomass pretreatment. During the bulrush pretreatment, supplementary $AlCl_3$ could be used to assist ChCl:Gly pretreatment, acquiring high delignification (70.8 %), xylan elimination (87.7 %), and the saccharification yield (90.1 %) [14]. Therefore, it deserves the development of new acidic DESs containing metal ion and surfactant for efficiently improving the enzymatic saccharification efficiency of lignocellulose.

During the pretreatment process in acidic systems, there is also the opportunity to produce high-value products xylo-oligosaccharides (XOSs), and furfural [15], which are both important biobased additives or intermediates in the food and pharmaceutical industries, with high application value [16]. It is recognized that acidic DES containing organic acids can promote the hydrolysis of hemicellulose, leading to the elevated saccharification capacity and acquiring byproducts such as xylose and XOSs. Additionally, metal chlorides, acting as Lewis acids, can favor the degradation of xylose and greatly accelerate the generation of furfural [17,18]. The protons provided by organic acids (such as lactic acid, LCA) can not only promote the efficient hydrolysis of hemicellulose and lignin, but also greatly inhibit the deactivation of metal ions due to hydration. Through the synergistic action of Brønsted acid and Lewis acid, the structure of lignocellulose is effectively disrupted, and enzymatic hydrolysis is enhanced to obtain a high yield of reducing sugar. Moreover, this synergy facilitates the hydrolysis of xylan and dehydration of xylose, promoting the formation of XOSs and furfural [15]. Supplementary of 0.2 mol/L formic acid into an aqueous system helped achieve the XOSs yield of 68 % and enhance biomass saccharification ability (160 °C, 120 min) [19]. Catalyst AT-Sn-YB immobilized with Lewis acid sites (Sn^{4+}) was used for the catalytic upgrading of bamboo, resulting in the production of 17.2 g/L fermentable sugars and 2.8 g/L XOSs, with an 80.3 % yield of furfural [20]. To accomplish high biomass saccharification and hemicellulose utilization, the organic acid and metal chloride-containing DESs were attempted for the synthesis of XOSs, furfural and reducing sugars.

To well understand the pretreatment mechanism, molecular dynamics (MD) simulations and quantum chemistry calculations can be applied to analyze the molecular structure of lignocellulose and explore deeply the interaction between the solvent and lignocellulose. Accordingly, in order to further investigate the interactions between C_nTAB/LCA/Fe^{3+} and lignocellulose at the molecular level, MD simulations and quantum chemical calculations can be attempted to deeply interpret the pretreatment mechanism combining with structural characterizations [21].

It is known that rapeseed straw (RS) represents a readily available bioenergy resource that is often disregarded and subjected to disposal or incineration practices [22,23]. It is of great interest to valorize RS into valuable chemicals through the effectual pretreatment. In this study, it was predicted that the C_nTAB-based DES pretreatment would significantly alter the physicochemical characteristics and microstructures of RS. Additionally, the increased accessibility of cellulose, and weakened hydrophobicity might be beneficial for enzymatic digestion of RS. Correspondingly, an effectual C_nTAB-based DES pretreatment process was established to address the recalcitrance of RS and mitigate the inhibitory effects of lignin and hemicellulose on enzymatic saccharification. In addition, the co-production of XOSs, furfural, and reducing sugars was developed through the efficient pretreatment. Initially, the chain length of C_nTAB (12–18), organic acid type, and metal salt type in C_nTAB-based DESs were systematically screened using single factor analysis, and the response surface method (RSM) was used to optimize the pretreatment conditions and further improve the saccharification effect of cellulose. Afterwards, the synergistic production potential of fermentable sugars, furfural, and XOSs within the pretreatment system was testified, considering factors (such as C_nTAB chain length, and $FeCl_3$ ratio) and their impact on pretreatment efficiency. Various analytical techniques such as FT-IR, CLSM, and SEM were employed to explore the effect of TTAB/LCA/Fe^{3+} on the RS morphology and properties. Molecular dynamics simulations were also utilized to gain insights into the changes in biomass properties at a molecular level. These investigations aimed to elucidate the mechanism for underlying the synergistic production and provided valuable information for the development of efficient biomass refining strategies through effectual pretreatment.

2. Materials and methods

2.1. Materials and chemicals

Rapeseed straw (RS) was from Taizhou city (Jiangsu, P.R. China), which was composed of 30.2 % of glucan, 15.1 % of xylan, 18.9 % of lignin, 16.2 % of ash, 5.5 % of crude protein and 11.4 % of water. The cellulase (Cellic® CTec2) was from Novozymes (Franklinton, North Carolina, U.S.A.). Cetyltrimethylammonium bromide, lactic acid, and others reagents were from Macklin Biochemical Co., Ltd. (Shanghai, P.R. China).

2.2. DES preparation and characterization

The C_nTAB-based ternary DESs were acquired by mixing metal chlorides, C_nTAB, and lactic acid (LCA) and stirring at 80 °C for 60 min. The viscosity of DESs was determined using a DV-1 T rotational viscometer from Fangrui Instruments Co., Ltd. (Shanghai, China). The surface tension of the DESs was measured using a BZT-2 automatic surface tension meter manufactured by Hengping Instruments Co., Ltd. (Shanghai, China). The absorption spectra of Nile red, *p*-nitroaniline, *N,N*-diethyl-*4*-nitrosoaniline and DES were determined at room temperature. and the Kamlet-Taft parameters were calculated based on the obtained spectra [24].

$$\alpha = \left(19.9657 - 1.0241 \times \pi^{*} - \frac{1}{\lambda_{max\,,\,A} \times 10^{-4}}\right) \div 1.6078 \quad (1)$$

$$\beta = 11.134 - 3058 \div \lambda_{max\,,\,B} - 1.125{*}\pi^{*} \quad (2)$$

$$\pi^{*} = 0.314 \times \left(27.52 - 1 \div \frac{1}{\lambda_{max\,,\,C} \times 10^{-4}}\right) \quad (3)$$

where $\lambda_{max\,,\,A}$, $\lambda_{max\,,\,B}$, and $\lambda_{max\,,\,C}$ stand for the maximum absorption wavelength of Nile Red, 4-nitroaniline, and *N,N*-diethyl-4-nitrosoaniline, respectively.

2.3. Pretreatment with TTAB/LCA/Fe^{3+}

The molar ratio of TTAB to LCA in DESs was fixed at 1:4, $FeCl_3$/TTAB value (molar ratio of $FeCl_3$ to TTAB) was 5:1–25:1 (mM:M). The pH was detected using a PHS-2F pH meter (Rex Electric Chemical, Shanghai, P. R. China). The TTAB/LCA/Fe^{3+} treatment parameters were optimized

(response surface methodology), such as reaction temperature (A), pretreatment time (B), and $FeCl_3$/TTAB value (molar ratio between $FeCl_3$ and TTAB) (C) on the glucose yield. Seventeen experimental groups and regression coefficients were calculated using a quadratic model. Pretreatments were conducted with a solid/liquid ratio of 1:10 at performance temperatures ranging from 120 to 180 °C for 20 to 90 min, using TTAB/LCA/Fe^{3+} with different $FeCl_3$/TTAB value (5:1–25:1, mM: M). Afterwards, the pretreated biomass was thoroughly cleaned with double distilled water (ddwater). The residue was dried in a 70 °C oven for 12 h and store at cool place. The biomass samples were enzymatically hydrolyzed by CTec.2 cellulase (15 FPU/g glucan) in citrate buffer (pH 4.8) at 50 °C for 72 h. The glucose yield and comprehensive severity factor was calculated using the following equations:

$$\mathrm{Log}_{R_0} = t \times e^{\left(\frac{T-100}{14.75}\right)} \quad (4)$$

$$\text{Comprehensive severity factor (CSF)} = \mathrm{Log}_{R_0} - \mathrm{pH} \quad (5)$$

$$\text{Glucose yield (\%)} = \frac{\text{Glucose generated from RS (g)}}{\text{Glucan in RS (g)} \div 0.9} \times 100 \quad (6)$$

In the above Eq. (4), t (min) stands for pretreatment time and T (°C) stands for temperature.

2.4. Chemical composition analysis

The chemical composition of biomass was analyzed as previously described [25]. The measurement of furfural and other sugars were performed using HPLC equipped with an Aminex HPX-87H column (Bio-Rad, USA). XOSs were calculated by measuring the xylose increment in 4 % sulfuric acid hydrolysis at 121 °C. The glucan recovery, xylan removal, and delignification in biomass were calculated as following equations:

$$\text{Delignification (\%)} = \left(1 - \frac{\text{Lignin in treated RS (g)}}{\text{Lignin in untreated RS (g)}}\right) \times 100 \quad (7)$$

$$\text{Xylan removal (\%)} = \left(1 - \frac{\text{Xylan in treated RS (g)}}{\text{Xylan in untreated RS (g)}}\right) \times 100 \quad (8)$$

$$\text{Glucan recovery (\%)} = \frac{\text{Glucan in treated RS (g)}}{\text{Glucan in untreated RS (g)}} \times 100 \quad (9)$$

$$\text{XOSs (g/L)} = (\text{Xylose after acidolysis (g/L)} - \text{Xylose in pretreatment hydrolysate (g/L)}) \times 0.88 \quad (10)$$

2.5. Regeneration of dissolved lignin from hydrolysate

After pretreatment, the filtered hydrolysate was added to 10 times volume of ddwater drop by drop to regenerate dissolved lignin. Gradual precipitation was conducted under strong magnetic stirring and left overnight. After centrifugation, the lignin was washed with acidic water (pH = 2.0) and dried for preservation.

2.6. Reuse of TTAB/LCA/Fe^{3+} in biomass pretreatment

In the re-utilization experiment of TTAB/LCA/Fe^{3+}, a solvent with a molar ratio of 1:4:0.0111 was used to pretreat RS at a solid-liquid mass ratio of 1:10 under the optimal pretreatment conditions. After each pretreatment, DES and biomass residue were separated by vacuum filtration while hot. The residue was washed several times for subsequent experiments, while the filtrated DESs were transferred to a flask for the next round of RS pretreatment.

2.7. Characterization of TTAB/LCA/Fe^{3+}-treated biomass

2.7.1. Hydrophobicity

Rose Bengal is a hydrophobic dye that readily adsorbs onto hydrophobic surfaces, such as lignin [26]. In this research, Rose Bengal dye adsorption was used to investigate the hydrophobicity of biomass, and the hydrophobicity (L/g) was calculated as previously reported [12].

2.7.2. SEM, CLSM and FT-IR

Scanning Electron Microscopy (SEM) analysis was performed at an acceleration voltage of 5.00 kv (JEOL JSM-6360LA, Japan). Confocal laser scanning microscopy (CLSM) (Nikon ECLIPSE Ti2, Japan) was employed to capture images, and NIS-Elements Viewer software was used for image processing. The structural changes during RS pretreatment were investigated utilizing Fourier Transform Infrared (FT-IR) (Thermo Fisher Nicolet IS50, U.S.A.) over 4000 to 400 cm^{-1}. The crystallinity ratio (CR) of biomass can be calculated through the formula below:

$$\text{Crystallinity ratio (CR, \%)} = \frac{A_{2900}}{A_{1372}} \times 100 \quad (11)$$

2.8. Quantum chemical calculations and molecular dynamics simulations

The interaction between TTAB/LCA/Fe^{3+} and lignocellulose was investigated, and models of the three main components of lignocellulose were determined based on previously described molecular dynamics simulations of lignocellulose [27,28]. The structure optimization of all molecules was implemented using the Gaussian09 software with the M062X/6-31G* base set. The Multiwfn program was utilized to calculate RESP charges [29]. Molecular dynamics simulations were implemented using GROMACS 2020 software using CHARMM force field files under periodic boundary conditions. The system was energy-minimized using the steepest descent method and then equilibrated under NPT ensemble at 435.7 K (162.5 °C) and 1 bar pressure for a 10 ns simulation. The trajectory data were subsequently analyzed and visualized using Visual Molecular Dynamics [30].

2.9. Energy flow

The higher heating values (HHVs) of xylan, glucan, and lignin present in RS were estimated using average HHV values obtained from experimental measurements of lignocellulosic biomass [31]. To determine the HHVs of the chemicals involved in this study, such as glucose, xylose, furfural, and XOSs, the Mendeleev formula was employed.

$$HHVs\ (MJ/kg) = 0.3491C + 1.254H - 0.1034(O - S) \quad (12)$$

$$\text{Heat values (MJ)} = \text{HHV (MJ/Kg)} \times \text{Mass of substances} \quad (13)$$

3. Results and discussion

3.1. DES screening

To comprehensively select the most suitable acidic *n*-alkyltrimethylammonium bromide-based (C_nTAB) DES for rapeseed straw (RS) treatment, the three components of the DES were screened using a single-factor method (160 °C, 60 min). Firstly, four C_nTAB with different

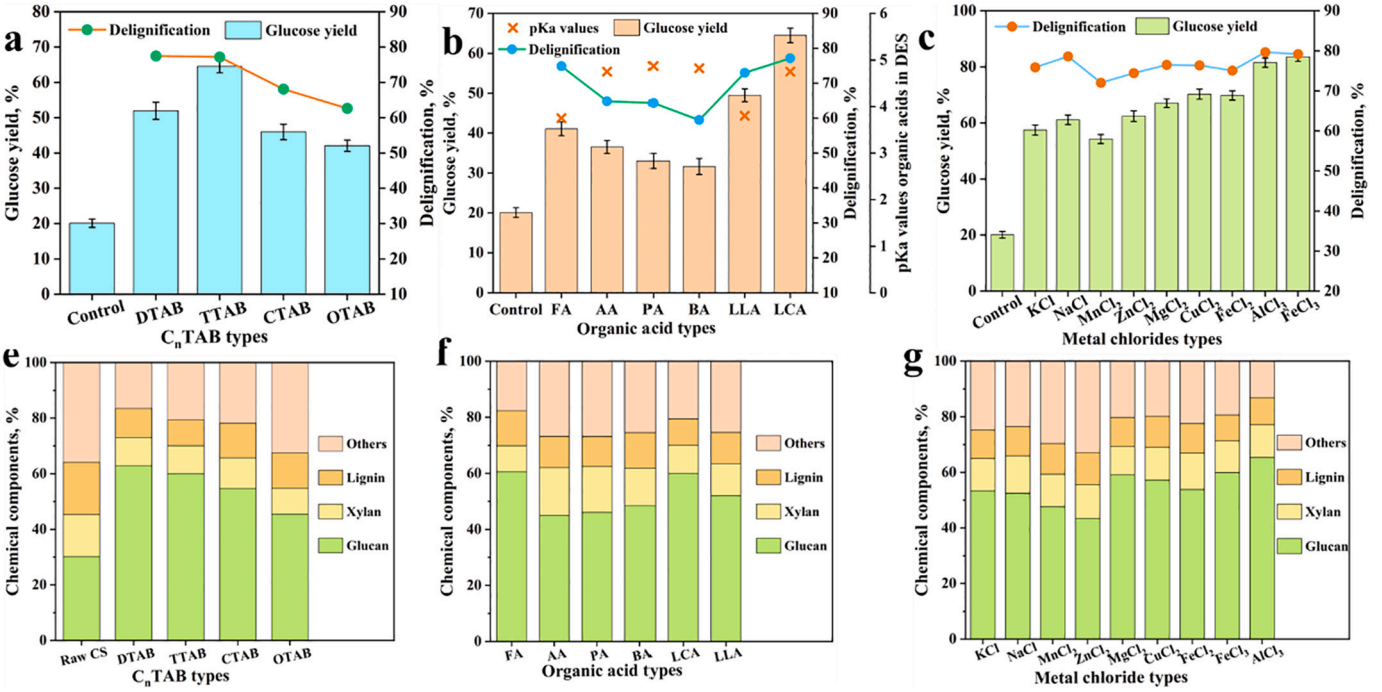


Fig. 1. The impact of different C_nTAB (a), organic acid (b) and metal chloride (c) in DES on glucose yield, and delignification; The chemical composition of RS pretreated with DES containing different C_nTAB (d) organic acids (e) and metal chlorides (f) [pretreatment conditions; 160 °C, 60 min, $FeCl_3$/TTAB = 5:1 (mM:M)].

alkyl chain lengths (12–18) were used to synthesize binary DESs with lactic acid (LCA) for pretreatment of RS. The impacts of different C_nTAB, organic acid and metal chloride in DES were examined on the changes of glucose yield and RS components (Fig. 1). Fig. 1a showcased that the glucose yield from biomass didn't increase consistently as the chain length of C_nTAB increased from 12 to 18. It manifested that the tetradecyltrimethylammonium bromide (TTAB)-based DES obtained the best pretreatment effectiveness, acquiring an enzymatic digestion efficiency of 64.5 %. As illustrated in Fig. 1e, it could be observed that xylan and lignin in the biomass was significant eliminated as the chain length of C_nTAB decreased, increasing the xylan elimination from 65.5 % to 73.4 % and delignification from 61.8 % to 77.4 % (Table S1, see supplementary materials). After pretreatment with DTAB/LCA, the solid recovery was <40 %, which was the lowest among all the C_nTAB-based DESs tested. It turned out that the DTAB resulted in the most significant structural disruption. However, although the dodecyltrimethylammonium bromide (DTAB) with the shortest alkyl chains exhibited the most significant structural disruption based on chemical composition, it didn't acquire the best glucose yield. This was likely due to the higher production of inhibitory compounds during the more intense pretreatment process caused by DTAB, thereby limiting the saccharification efficiency of RS [32].

The Brønsted acid in the ternary DES played a crucial role in biomass pretreatment by providing a large number of protons for attacking the chemical bonds in biomass, thus substantially reducing the recalcitrance of lignocellulosic materials [19]. In this research, four different monocarboxylic acids with varying chain lengths and two common organic acids (lactic acid and levulinic acid) were individually mixed with the best C_nTAB obtained above, and TTAB-based DESs with different organic acids were screened to acquire the most suitable binary C_nTAB-based DES for RS pretreatment. From Fig. 1f, the RS treated with DES containing lactic acid (LCA) had the lowest lignin content (9.4 %). It could be calculated that DES containing LCA effectually eliminated the highest amount of lignin (77.2 %) and xylan (69.7 %) within the system, followed by the DES with formic acid, reaching 74.9 % and 76.8 %, respectively. Among these organic acids, butyric acid exhibited the weakest disruption of the lignocellulose structure, which might be attributed to its relatively higher pKa value. Fig. 1b indicated that the RS pretreatment using C_nTAB-based DES containing LCA exhibited the best performance, with an enzymatic saccharification efficiency of 64.5 %. Additionally, it was observed that the elimination capacity of monocarboxylic acids for lignin and xylan gradually weakened with increasing chain length. This trend was closely related to the pKa values, as showcased in Fig. 1b. As the chain length of monocarboxylic acids increased, their pKa values also rose, resulting in a decline in pretreatment efficiency. However, the pKa of acids couldn't be used directly to predict pretreatment results. When organic acids such as lactic acid possess more complex structures, including hydroxyl and carbonyl groups, the correlation between pKa and pretreatment performance couldn't be simply described as negative. On one hand, different functional groups apparently influenced the stereostructure of organic acids, thereby affecting the pretreatment process. On the other hand, complex organic acids might exhibit intramolecular interactions, such as the significant impact of hydroxyl groups on hydrogen bonding or hydrophobic interactions, which would influence the ionization equilibrium of carboxyl groups.

In the pretreatment system, metal chlorides can not only act as Lewis acid to catalyze the hydrolysis of chemical bonds such as β-1,4-glucoside and β-O-4 bonds in hemicelluloses and lignin [33], but also react with hydroxyl and phenolic groups in lignocellulose to form metal ion-lignin complexes, making them easier to degrade [13]. In Fig. 1c, the pretreatment effectiveness of nine DESs containing different metal chlorides was compared. It was manifested that the ternary DESs containing metal chlorides exhibited apparently improved pretreatment performance compared to TTAB/LCA. The lowest enzymatic digestion efficiency for RS pretreatment with TTAB/LCA/Mn^{2+} was 54.3 %, while higher saccharification efficiencies were acquired with TTAB/LCA/Fe^{3+} and TTAB/LCA/Al^{3+} treatments, reaching 83.6 % and 81.5 %, respectively. Considering the comprehensive screening of components in the ternary DES above, TTAB/LCA/Fe^{3+} was considered the optimal C_nTAB-based ternary DES for RS pretreatment.

It is widely recognized that the properties of DESs have substantial impacts on the elimination of biomass components [34]. As shown in the FT-IR image of TTAB/LCA/Fe^{3+} (Fig. S1, see supplementary materials),

a peak due to the stretching vibration of -OH was observed at 3450 cm^{-1}, showing a blue shift compared to LCA, indicating the formation of a new hydrogen bond network in DES [35]. The peak appeared at 1746 cm^{-1} was associated with C═O in from LCA [36]. A peak at 966 cm^{-1} was attributed to the C—N group of TTAB and the absorption around 744 cm^{-1} was connected with Fe^{3+}-OH- Fe^{3+} [37]. In biomass pretreatment, solvent viscosity plays a crucial role in the diffusion rate of molecules [34]. There was a substantial difference in viscosity among DESs synthesized using C_nTAB with different chain lengths (Table S2, see supplementary materials). It was evident that DESs synthesized with shorter alkyl chain lengths exhibit lower viscosity. The viscosity of octadecyltrimethylammonium bromide (OTAB) was 1983.5 cp, while that of DTAB was only 350.3 cp. It was probably related to lower intermolecular friction caused by lower chain length, thus resulting in better mobility. This helped the molecules move more easily in the solvent, confirming that TTAB/LCA/Fe^{3+} had better pretreatment performance. Additionally, at room temperature, ternary DESs showed a significant reduction in viscosity compared to binary DESs. The viscosity of TTAB/LCA reached 848.2 cp, whereas TTAB/LCA/Fe^{3+} had only 674.2 cp. These findings regarding viscosity were consistent with previous studies on the pretreatment effectiveness of different DESs.

Surface tension is also a crucial solvent property known to influence the pretreatment process. Solvents with low surface tension might better interact with biomass and rapidly penetrate structures such as lignin. The formed stronger interactions would facilitate their efficient removal. TTAB/LCA/Fe^{3+} had a surface tension of only 30.2 mN/m (Table S2, see supplementary materials). In comparison, the surface tension of water was 72 mN/m (25 °C), significantly higher than that of this efficient pretreatment solvent. Consistent with the viscosity trend, the shorter chain lengths of C_nTAB also contributed to lower surface tension in DESs, thereby promoting DES penetration into lignocellulose structures and accelerating the deconstruction of lignin and other components. The Kamlet-Taft (K-T) parameters, including polarity (π*), acidity (α), and basicity (β), are often used to describe the solvents and assess their solubility and dissolution behavior [38,39]. Firstly, it was found that there were certain trends in K-T parameters among DESs synthesized using C_nTAB with different chain lengths. When the alkyl chain length increased from 12 to 18, the α value of DES decreased from 1.90 to 1.02, while the β value decreased from 1.65 to 1.39, indicating a decrease in the ability of providing or receiving hydrogen bonds. Additionally, C_nTAB-based binary DESs exhibit low polarity, with π* values all below 0.1. This was related to the use of C_nTAB, which had low polarity, as a hydrogen-bond-acceptor in DES synthesis. In the presence of metal chlorides, it was manifested that the hydrogen-bond acidity and basicity of ternary DESs were elevated. The α value of TTAB/LCA/Fe^{3+} reached 1.87, and the β value increased to 2.08. Moreover, an increase in π* value was observed, indicating a significant enhancement in solvent polarization due to the introduction of metal ions [40] (Table S2, see supplementary materials). It was likely that the introduction of $FeCl_3$ as an HBA in TTAB/LCA/Fe^{3+} strengthened the hydrogen-bond network, altered DES physicochemical properties, and enhanced the capability to better deconstruct lignocellulose.

3.2. *Investigation of TTAB/LCA/Fe^{3+} treatment temperature, duration and $FeCl_3$/TTAB ratio on the glucose yield*

To build an efficient TTAB/LCA/Fe^{3+} pretreatment system, the response surface methodology (RSM) using Box-Behnken design was applied to optimize the pretreatment conditions [A: reaction temperature (120–180 °C), B: reaction time (20–90 min), C: $FeCl_3$/TTAB value (5:1–25:1, mM:M)]. A total of 17 optimization experiments were conducted using the novel ternary DES TTAB/LCA/Fe^{3+} and the chemical compositions were listed in Table S3 (see supplementary materials). The experimental data were analyzed using Design Expert 13, with glucose yield as the response value, resulting in regression equations that described the relationship between the different pretreatment

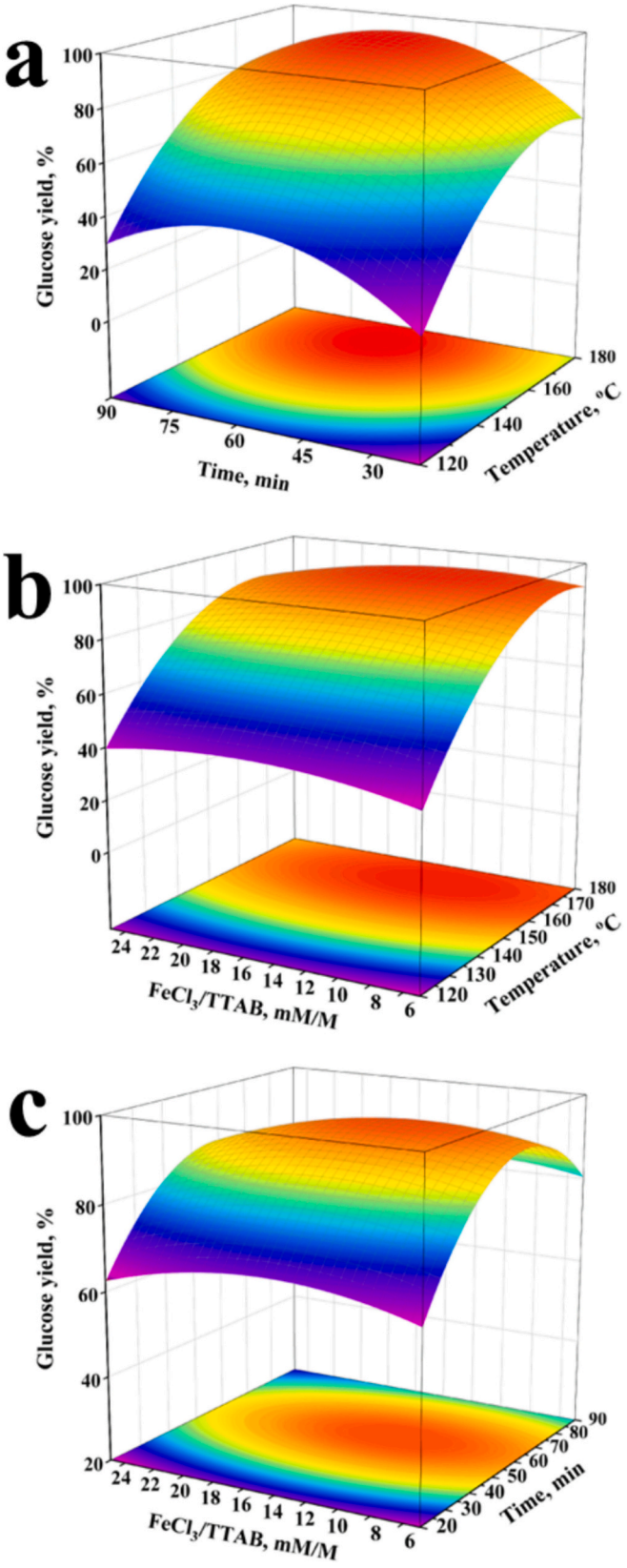


Fig. 2. The optimization of treatment temperature (120–180 °C) and treatment duration (20–90 min) on the glucose yield (a); The temperature (120–180 °C) and $FeCl_3$/TTAB value (5:1–25:1, mM:M) of TTAB/LCA/Fe^{3+} on the glucose yield (b); The time (20–90 min) and $FeCl_3$/TTAB value (5:1–25:1, mM:M) of TTAB/LCA/Fe^{3+} on the glucose yield (c) in RS pretreatment by TTAB/LCA/Fe^{3+}.

parameters and enzymatic digestion efficiency.

As showcased in the temperature/time two-factor analysis (Fig. 2a), when the pretreatment time was fixed, an increase in temperature (120–150 °C) led to an elevated enzymatic digestion efficiency. However, further increasing temperature didn't substantially promote enzymatic digestion efficiency and even resulted in a slight decrease. This phenomenon was attributed to the accelerated molecular motion at higher treatment temperatures, which promoted the hydrolysis reaction [41]. However, extreme temperatures also catalyzed the generation of hydrolysis inhibitors [42]. The condensation reactions between various free furanic compounds and aromatic compounds in the system were accelerated, leading to the formation of humins and pseudo-lignin. These by-products covered the cellulose surface and hindered the binding of cellulase to catalytic sites, ultimately resulting in declined enzymatic digestion capacity [43].

In Fig. 2b and c, the influence of two factors, temperature/time and $FeCl_3$/TTAB value, on enzymatic digestion capacity was comprehensively presented. When the pretreatment duration or temperature was fixed, the enzymatic digestion efficiency of cellulose substrate initially elevated and then dropped with an increase in the $FeCl_3$/TTAB value. When RS was pretreated at 162.5 °C for 61.7 min, and $FeCl_3$/TTAB was 11.1:1 (mM:M), the predicted enzymatic hydrolysis efficiency reached its highest value. This efficiency was about 94 %. In the TTAB/LCA/Fe^{3+} system, $FeCl_3$ might act as a hydrogen bond acceptor (HBA) and interact with TTAB and lactic acid, forming complex metal cations [15]. During the pretreatment of RS, LCA as Brønsted acid and $FeCl_3$ as Lewis acids acted synergistically, disrupting the covalent bonds in hemicellulose and lignin. This elimination of physically compacted barriers on the cellulose surface substantially enhanced cellulose accessibility. However, the existence of a large amount of $FeCl_3$ also created conditions for the generation of inhibitors. During the pretreatment process, polymers in biomass like lignin and hemicellulose tended to break down into small molecular compounds (such as lignin monomers and furfural). These freely present small molecules underwent a series of cross-linking and condensation reactions under harsh pretreatment conditions (e.g., excessive $FeCl_3$ existed in the system), leading to the formation of new covalent bonds and the waste of carbohydrates [44].

Due to the multitude of factors influencing pretreatment effectiveness, a more comprehensive indicator is needed to assess the intensity of pretreatment [45]. The comprehensive severity factor (CSF) is a parameter for better understanding of the influence of reaction conditions and chemical reactions in biomass pretreatment [45]. The relationship involving delignification, xylan elimination and CSF was showcased in Fig. 3a. Most data points were encompassed within the 95 % confidence interval, and the R-squared values for the two fitted equations reached 0.93 and 0.95, respectively. These results indicated that the Boltzmann model could be used to accurately express and even predict the relationship involving delignification, xylan elimination, and CSF. Increasing CSF might cause the enhanced elimination of lignin and xylan. However, it was worth noting that this growth trend was nonlinear. As CSF increased from 2.93, the enhancement of lignin and xylan elimination was slow at first, and then gradually accelerated. When CSF exceeded about 4.5, the rate of enhancement started to slow down. This was likely due to the enhanced interaction and reaction activity between the DES and cellulose as CSF increased, leading to an accelerated pretreatment effect. As CSF further elevated, the growth of the reaction rate would become stable after the temperature reached a critical point, and the dissolution capacity of the solvent might be saturated [45].

Through ANOVA analysis, the model employed in this study had an F-value of 34.65 with a p-value <0.01, and the lack-of-fit term was found to be non-significant (Table S4, see supplementary materials). Consequently, the obtained model in this research manifested excellent significance and statistical relevance. According to the quadratic polynomial model (Eq. 13), the optimal temperature, time, and $FeCl_3$/TTAB value (mM:M) for pretreating RS with TTAB/LCA/Fe^{3+} were

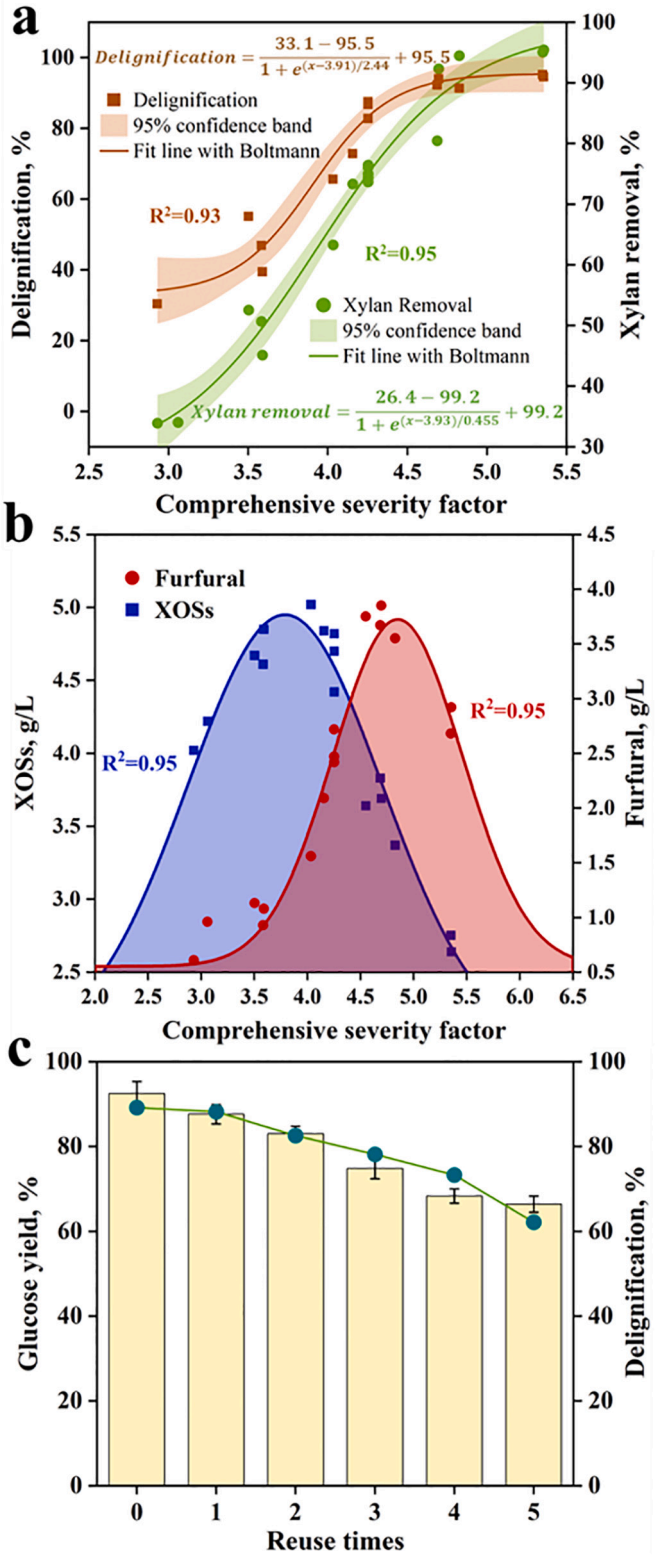


Fig. 3. The correlations between comprehensive severity factor (CSF) with delignification and xylan elimination (Boltzmann model) (a); The correlations of CSF with furfural and XOSs content (Gauss model) (b); The effect of DES reuse on glucose yield and delignification in RS pretreatment by TTAB/LCA/Fe^{3+} (c).

determined to be 162.5 °C, 61.7 min, and 11.1:1, respectively. The highest enzymatic digestion efficiency (94.2 %) was acquired under the predicted optimal pretreatment conditions. To validate the accuracy of this quadratic polynomial model, the predicted optimal pretreatment conditions were replicated to acquire the actual enzymatic saccharification efficiency. After performing three parallel experiments, the average saccharification efficiency was acquired to be 92.5 %, which closely matched the predicted value. It manifested its remarkable reliability. Through the treatment under optimal conditions, 89.2 % of lignin and 77.9 % of xylan in RS were removed, and about 80 % of glucan retained. These results implied that the optimal conditions for pretreating RS using TTAB/LCA/Fe^{3+} (with the $FeCl_3$/TTAB value of 11.1:1, mM:M) were at 162.5 °C for 61.7 min, resulting in a maximum enzymatic saccharification efficiency. The glucose yield was fitted as follows:

$$\begin{aligned}\text{Glucose yield (\%)} =& 0.8662 + 0.2486 \times A + 0.0647 \times B - 0.0067 \times C \\ &+ 0.0009 \times A \times B - 0.0268 \times A \times C - 0.0183 \times B \\ &\times C - 0.1899 \times A^2 - 0.1961 \times B^2 - 0.0576 \times C^2\end{aligned} \tag{14}$$

After the pretreatment, the dissolved lignin was also precipitated for analysis to understand the structure of the dissolved lignin (Fig. S2, see supplementary materials), providing new evidence for further analysis of the pretreatment mechanism. The unconjugated carbonyl stretching present at 1730 cm^{-1} was attributed to the C═O group and the band at 1631 cm^{-1} was due to the stretching of conjugated carbonyls in lignin. The aromatic framework vibrations of the lignin components were exhibited at 1462 and 1432 cm^{-1} [46]. The peak at 1370 cm^{-1} was attributed to non-etherified phenol-OH caused by cleavage of the α-O-4 and β-O-4 bonds, confirming the acid-catalyzed hydrolysis of lignin by DES during pretreatment [47]. The bands located at 1168 cm^{-1}, 1117 cm^{-1}, and 1057 cm^{-1} of regenerated lignin were attributed to the typical structures of hydroxyphenylpropane, guaiacyl, and syringyl in lignin [48].

Reusability is undoubtedly an important property of an effectual pretreatment solvent [15]. In this research, TTAB/LCA/Fe^{3+} was reused five times to observe its performance in lignin elimination and other aspects (Table S5, see supplementary materials). During each batch of repeated use, a part of the DES was lost, which usually occurred in the vacuum filtration step. Based on the weight of solvent before and after recycling, about 80–90 % of the solvent could be recovered per reuse. After the pretreatment, the biomass residue was separated, and the remaining TTAB/LCA/Fe^{3+} was reused. Upon the third reuse of the solvent, the xylan elimination declined to 65.7 %, while the lignin elimination dropped to 78.2 % (Table S5, see supplementary materials). Accordingly, the solvent manifested an apparent effect on eliminating the structures for inhibiting enzymatic hydrolysis of biomass even after multiple reuses. After five cycles of reuse, the elimination of lignin and xylan apparently declined to 62.1 % and 56.0 %, respectively (Table S5, see supplementary materials). This decrease could be attributed to the degradation of components within DES at high temperatures, leading to a reduced ability to dissolve biomass components. Additionally, in each pretreatment process, the degradation of lignin and the accumulation of other inhibitors may also lead to the decline of pretreatment performance. Consequently, the enzymatic saccharification ability of cellulose gradually diminished with each reuse. In this work, when the pretreatment efficiency of the solvent was significantly higher than that of untreated rapeseed straw (20.1 %), it could be considered to be reusable. It was observed that after three cycles of reuse, the enzymatic saccharification efficiency declined from 92.5 % to 74.1 %. After the fifth reuse, the saccharification efficiency of RS treated with TTAB/LCA/Fe^{3+} remained at 66.4 %, indicating the solvent's good reusability. In some published works, choline chloride (ChCl) and lactic acid (LCA) were used to synthesize a DES for the pretreatment of poplar, achieving a saccharification efficiency of 90.6 %, which still reached 51.5 % after four reuses [49]. Switchgrass was pretreated with acidified ChCl:Gly, yielding a glucose yield of 88.9 %, which dropped to 32.6 % after three reuses [50].

3.3. Investigation of TTAB/LCA/Fe^{3+} comprehensive severity factor and reused DES on the component elimination and carbohydrate conversion

Under the acidic environment, xylan can not only be hydrolyzed into a large amount of pentose units [45], but also be converted into xylo-oligosaccharides (XOSs) and furfural [20,51]. Under optimal pretreatment conditions, 3.75 g/L furfural and 3.64 g/L XOSs produced in this pretreatment liquor. As showcased in Fig. 3b, when the CSF was low, only a small amount of furfural was present in the solution due to a lower temperature. In fact, measurements revealed that almost no furfural was detected when the pretreatment temperature was 120 °C. When the CSF increased from 3.0 to 4.8, various reactions such as dehydration and isomerization of xylan-derived pentose units were promoted, leading to the synthesis of furfural and reaching a peak concentration of 3.85 g/L. As the CSF further elevated, a decrease in furfural content was observed, which was likely due to the occurrence of other side-reactions of furfural under extreme pretreatment conditions and acidic environment, leading to further condensation into humins or the generation of other byproducts [45].

During the TTAB/LCA/Fe^{3+} pretreatment, XOSs underwent a similar process as furfural. When the CSF was <3.5, the content of XOSs rapidly increased to approximately 5 g/L. However, as the solution environment became more severe, the content of XOSs decreased. The hydrolysis of the glycosidic bonds in its structure was accelerated, tending to degrade into its structural unit xylose or further dehydration to form furfural. However, the relationship between furfural and XOSs was not simply an antagonistic one. It was found that the rates of their side-reactions both increased under harsh treatment conditions, leading to their degradation [20]. To find mathematical models behind the data and better understand and predict the data points, a Gaussian model was used in this research to explore the relationship between CSF and furfural and XOSs yields. It could be seen that the fitted Gaussian equations well explained the relationship between CSF and the two high-value products, with R-squared values both reaching 0.95.

In addition to CSF, there also appeared to be a correlation between the C_nTAB types in ternary DES and the formation of furfural and XOSs. C_nTAB with different alkyl chain lengths (12–18) and lactic acid were used to synthesize DESs, and the pretreatment reaction was implemented under optimal conditions to detect the content of XOSs and furfural in the solution. Four different C_nTAB with different alkyl chain lengths (12, 14, 16, and 18) were utilized for synthesizing binary DESs (Fig. 4a). The alkyl chain length of TTAB was 14. By using TTAB/LCA as pretreatment solvent, the contents of XOSs and furfural were 2.25 g/L and 2.17 g/L, respectively. As the alkyl chain length of C_nTAB increased, the furfural content in the system decreased, while the XOSs content increased. This phenomenon was closely related to the non-cellulose component removal performance of different C_nTAB based DESs described above. It manifested that DESs formed by C_nTAB with longer alkyl chains exhibited relatively weaker xylan elimination performance, suggesting its low efficiency in the appreciation of hemicellulose.

As the $FeCl_3$/TTAB value of TTAB/LCA/Fe^{3+} elevated from 5:1 to 15:1 (mM:M), the furfural content gradually increased (Fig. 4b). When it further increased, the furfural content in the system no longer continued to rise. As the $FeCl_3$/TTAB value was over 15:1 (mM:M), the furfural content dropped. In addition, it was observed that XOSs seemed to be more sensitive to $FeCl_3$. The content of XOSs rapidly decreased when the $FeCl_3$/TTAB value increased. The phenomenon of rapid decrease in the content of both high-value products with the increase in $FeCl_3$/TTAB value was closely related to the catalysis of Lewis acid in the pretreatment. In a concise summary, the high-efficiency co-production of furfural and XOSs could be achieved under the optimal pretreatment conditions, and the carbohydrates in lignocellulose were

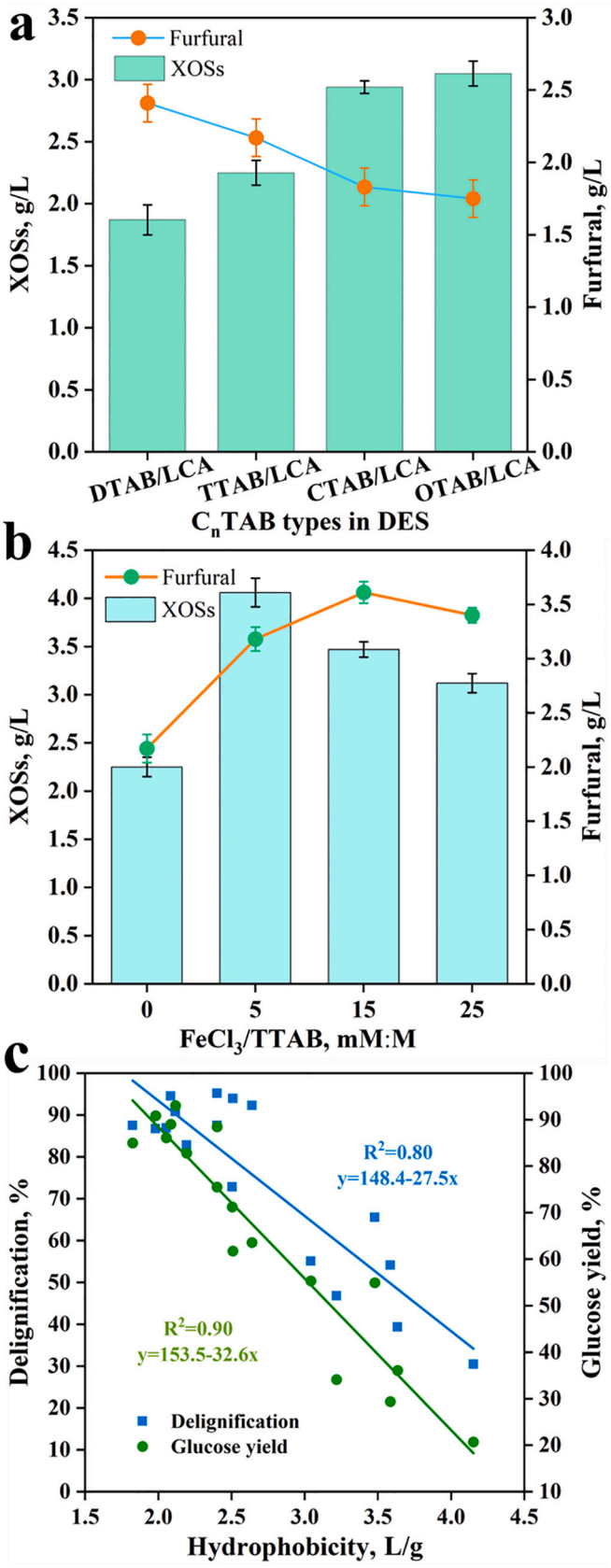


Fig. 4. The effect of C_nTAB types in DES on furfural and concentration; the effect of $FeCl_3$/TTAB value (5:1–25:1, mM:M) of DES on furfural and XOSs concentration under optimal pretreatment conditions (b). A linear fitting between delignification and glucose yield of RS and biomass hydrophobicity (c).

comprehensively upgraded. As a matter of fact, different polymerization degrees of XOSs have different applications in health products, pharmaceuticals, and other fields [52].

Evaluating the effectiveness of pretreatment methods for lignocellulose requires a comprehensive understanding of changes in biomass hydrophobicity [12]. In this research, the relationship between hydrophobicity, delignification and enzymatic hydrolysis capacity of RS was testified. The hydrophobicity of untreated RS was found to be 4.6 m^2/g. However, after TTAB/LCA/Fe^{3+} treatment (the optimal condition obtained above), the hydrophobicity substantially declined to 2.1 m^2/g. Concurrently, a strong linear relationship was observed between hydrophobicity and lignin elimination efficiency, with a fitted equation of 0.80 (Fig. 4c). The weakened hydrophobicity was attributed to the enhanced pretreatment process, which led to substantial disruption of the lignocellulosic structure in the biomass, resulting in the elimination of hydrophobic lignin [12]. Furthermore, the relationship between hydrophobicity and enzymatic hydrolysis capacity was described by a fitted linear equation ($R^2 = 0.9$). It was noteworthy that the decrease in hydrophobicity caused by the reduction of lignin played a crucial role in improving enzymatic hydrolysis ability [12].

3.4. Investigation of structural alteration of RS before and after pretreatment

FT-IR plays a crucial role in the analysis of biomass components and structures [53]. As showcased in FT-IR image (Fig. 5), 2914 cm^{-1} absorption peak corresponding to the cellulose structure maintained a high strength, implying that the cellulose matrix was preserved [54]. It was worth noting that the absorption peaks at 2851 and 1251 cm^{-1} probably were attributed to the -CH_3 and C—O groups in hemicellulose [55,56]. These peaks gradually diminished with an increase in CSF, which was associated with the significant elimination of hemicellulose during the pretreatment steps. The peak at 1618 cm^{-1} was attributed to carbonyl groups present in the biomass [57]. It was observed that this peak gradually weakened with harsher pretreatment conditions, which was speculated to be associated with the elimination or modification of the associated structures in lignin and hemicellulose. The absorption peak at 1750 cm^{-1} was attributed to non-conjugated carbonyl/carboxyl group stretching in the lignin structure [58]. The absorption peak near 1100 cm^{-1} and 819 cm^{-1} was associated with C—O stretching and glycosidic linkages [12,59] (Fig. 5).

Crystallinity ratio (CR) is an important parameter calculated through

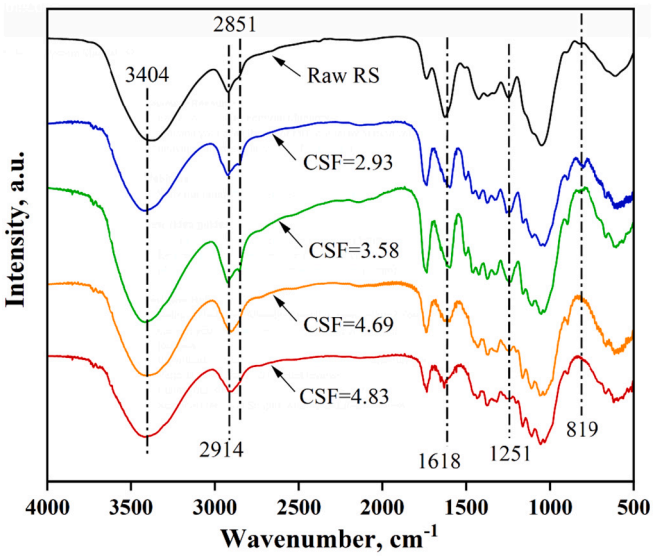


Fig. 5. The FT-IR image of raw RS and TTAB/LCA/Fe^{3+} treated RS with different CSF.

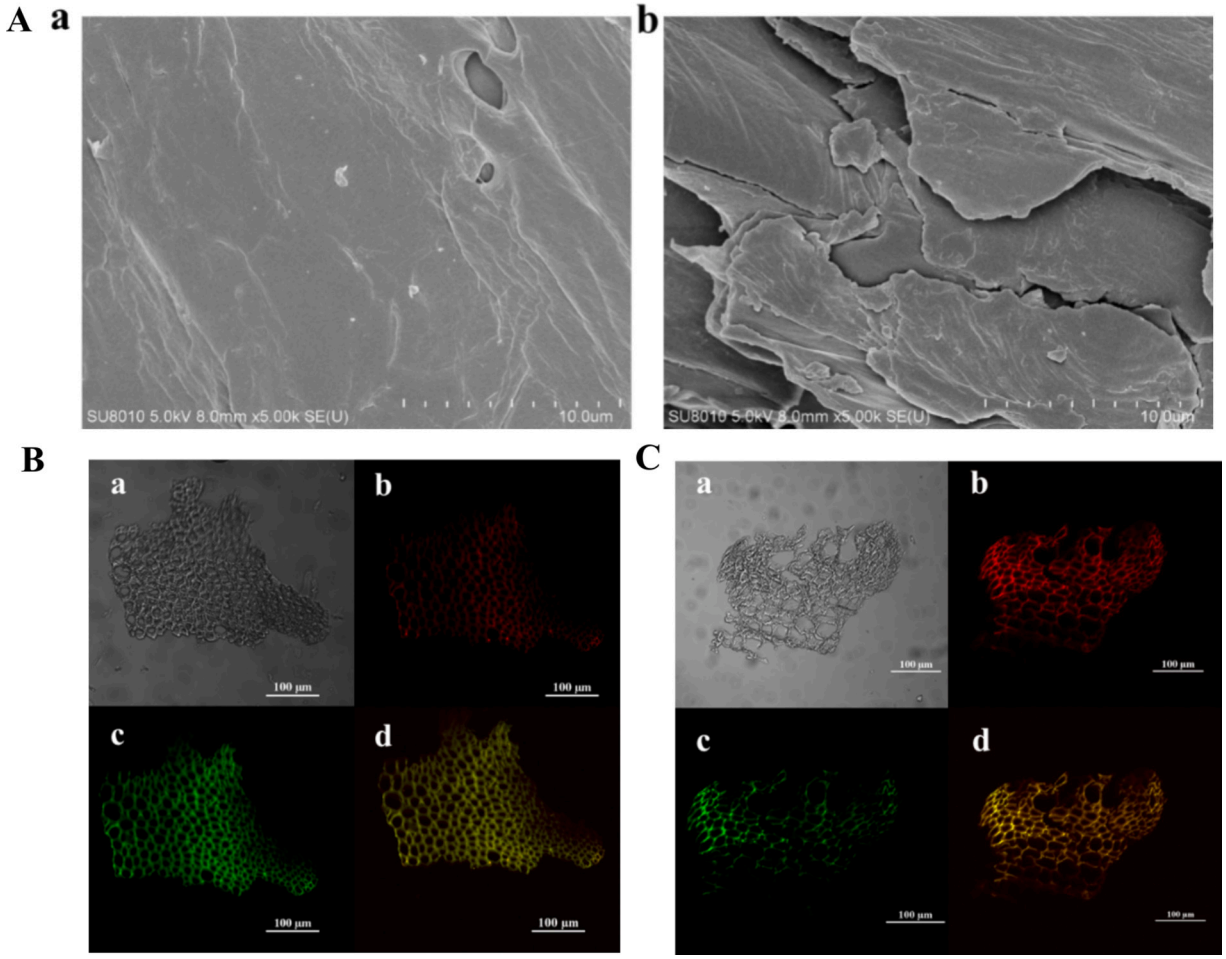


Fig. 6. The SEM image (A) [raw (a); TTAB/LCA/Fe^{3+} treated RS (b)]; The CLSM image of raw (B); TTAB/LCA/Fe^{3+} (C) treated RS [Brightfield image (a); cellulose (red) (b); lignin (green) (c); mixture of cellulose and lignin image (merged color) (d)].

FT-IR intensity [60]. After calculating the CR of the raw and treated samples, an increase in CR was observed. The CR reached 1.2 when the CSF was 4.83, while the CR of the raw RS was 1.02. The increase in CR further supported the elimination of lignin and hemicellulose, as these components helped to reduce CR values. Overall, the TTAB/LCA/Fe^{3+} treatment was effective in eliminating lignin and hemicellulose and increasing cellulose content, resulting in the higher enzymatic hydrolysis capacity.

SEM is a commonly used characterization technique to obtain the surface morphology of biomass materials [61]. The surface of RS after TTAB/LCA/Fe^{3+} treatment was rough, with large and irregular pores (Fig. 6A). In comparison, the untreated RS surface was smooth and had limited sites available for cellulase attachment. These surface morphology alterations were attributed to the disruption of cellulose and the substantial elimination of lignin and hemicellulose [61]. The combined effect of these alterations substantially enhanced the enzymatic hydrolysis capacity of biomass.

CLSM is a high-resolution microscopy technique that utilizes laser beams and an optical lens system to acquire three-dimensional, high-contrast images of samples [62]. In the CLSM images, green color represents lignin, and red color represents cellulose. It could be observed that the untreated RS exhibited intact cell structures, with clearly visible cells and smooth cell edges (Fig. 6B). However, during microscopic observation, it was difficult to find intact RS cells in samples treated with TTAB/LCA/Fe^{3+} due to the solvent-induced disruption of the lignocellulosic structure, as the cells appeared fragmented and scattered. By examining the color variations in the images, it was noted that the CLSM image of untreated RS appeared bright green, with an overall yellowish hue in the overlaid color images. This brightness was attributed to the presence of a large number of aromatic compounds and conjugated structures in lignin, which exhibited strong fluorescence. In contrast, the treated RS manifested a noticeable deepening of red color and a significant dimming of green color, implying a substantial elimination of lignin and an increase in cellulose content, consistent with the results of biomass composition measurements (Fig. 6C). In a concise summary, the CLSM images afforded clear evidence of the chemical and structural changes induced by TTAB/LCA/Fe^{3+} treatment in the biomass, corroborating the findings of previous component analyses and rendering the evidence for the enhanced saccharification.

3.5. *Molecular-level understanding of TTAB/LCA/Fe^{3+} treatment*

A molecular-level understanding of the interactions of lignocellulosic biomass can significantly contribute to the development of biomass refining industries and enhance environmental sustainability [63]. To investigate the properties of lignocellulosic biomass, atomistic density functional theory (DFT) calculations were employed to explore the interactions between TTAB/LCA/Fe^{3+} and lignocellulosic biomass models [64]. The obtained calculations afforded valuable insights into the configuration and the electrostatic potential (ESP) of DES components and three lignocellulosic biomass models (Fig. 7A).

The ESP images offered detailed information about the charge distribution within the biomass components. Specifically, hydroxyl hydrogen atoms in lignocellulosic biomass exhibited positive ESP, while

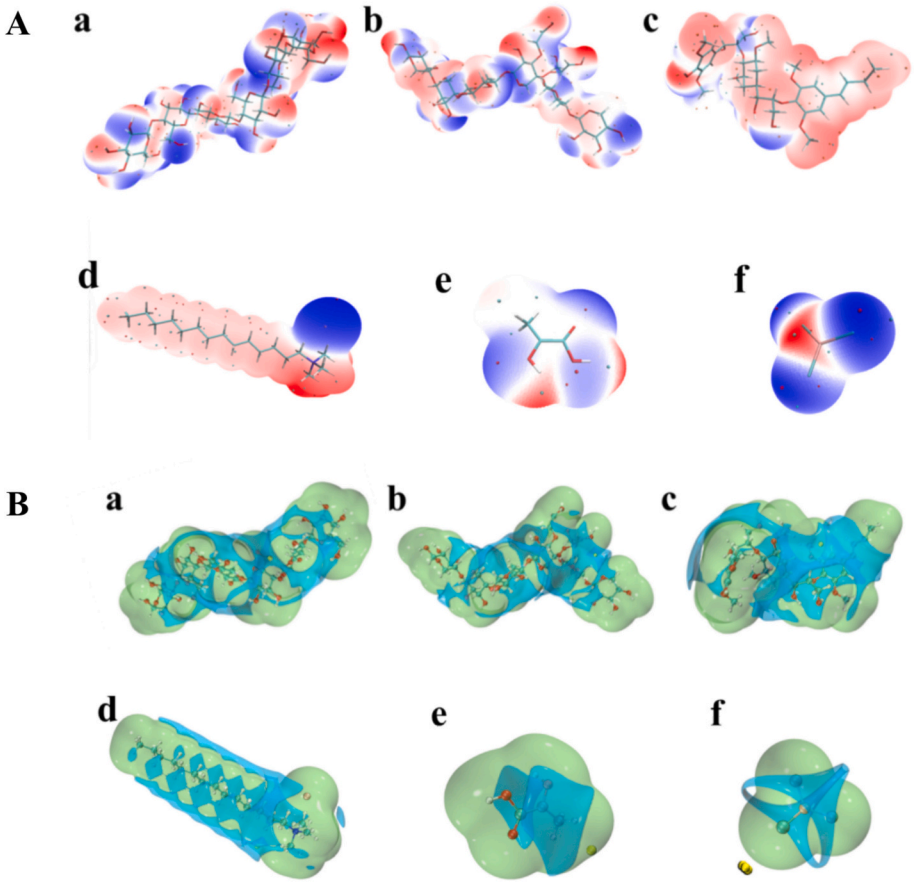


Fig. 7. The electrostatic (A) and van der Waals potential (B) cellulose model (a), hemicellulose model (b), lignin model (c), TTAB (d), LCA (e), and $FeCl_3$ (f).

hydroxyl oxygen atoms showed negative [65]. This charge distribution played a crucial role in the interactions between DES and biomass components. It was noteworthy that TTAB, especially its bromine component, introduced a negative electrostatic potential into the system and disrupted hydrogen bonding in lignin and hemicellulose. Ultimately, this disruption facilitated the removal of lignin and hemicellulose in biomass. In addition to electrostatic interactions, van der Waals forces also had a significant influence on the interactions between DES and biomass components, promoting interactions with non-polar compounds and aiding in the dissolution and extraction processes. In van der Waals potential images, the blue region represented the area where the van der Waals interaction appears as an attraction. (Fig. 7B). Clearly, it was observed that the area near the long alkyl tail of TTAB showed strong attraction. The occurrence of these interactions effectually facilitated the elimination of non-polar substances (such as lignin), making it particularly relevant in the delignification process using the TTAB/LCA/Fe^{3+} system. In addition, within the lignocellulosic biomass model, the lignin model exhibited the strongest dispersion interaction with the solvent, reaching a minimum potential energy of −0.93.

To further understand the interactions within the DES system, an independent gradient model (IGM) analysis was performed on the MD simulation system, revealing weak interactions between cellulose, hemicellulose, lignin models, and TTAB/LCA/Fe^{3+} (Fig. S3, see supplementary materials) [66]. Hydrogen bonding is a relatively strong non-covalent interaction that typically occurs between hydrogen on a strongly polar bond and an atom with a lone pair of electrons. Hemicellulose, being a polysaccharide rich in hydroxyl groups, could act as both a hydrogen bond donor and acceptor. As showcased in Fig. 8b, a large and dark blue region was observed and the blue scatter points had the highest density between hemicellulose and DES, implying strong hydrogen bonding interactions between them. Gromacs software was used to calculate the intermolecular energy of the molecular dynamics simulation process from 500 ps to 10 ns, and the electrostatic energy (E_{elec}) between hemicellulose and TTAB/LCA/Fe^{3+} was calculated to be −429.3 kJ/mol, the largest among the three models (Table S6, see supplementary materials). In fact, hydrogen-bond interaction was the main form of intermolecular electric interaction, and there was a direct correlation between them, which confirmed the stronger hydrogen bond interaction between the hemicellulose model and the DES. These robust hydrogen bonds played a promoting role in the hydrolysis process of hemicellulose, aiding in the disruption of covalent bonds between sugar molecules and facilitating the generation of XOSs and furfural. The structure of lignin consists of rigid aromatic rings connected by aryl ether bonds and flexible side chains through carbon-carbon bonds, lacking strongly polar side chain groups. Consequently, the interaction types between lignin and solvents were mainly van der Waals forces instead of hydrogen bond interaction. As shown in Fig. 8c, large green areas were observed between lignin and DES, which stands for van der Waals forces. Based on intermolecular energy calculations, the lignin model is the only one in which the van der Waals (E_{vdW}) energy (−269.46 kJ/mol) is greater than the electrostatic energy (−171.61 kJ/mol) (Table S6, see supplementary materials). Therefore, the main force between TTAB/LCA/Fe^{3+} and TTAB/LCA/Fe^{3+} is the van der Waals interaction, which is consistent with its hydrophobic structure [21]. This explained the efficient elimination and dissolution of lignin from biomass in the TTAB/LCA/Fe^{3+} system. Accordingly, the DFT calculations, electrostatic potential analysis, and IGM calculations following MD simulations have provided valuable insights into the properties of lignocellulosic biomass and its interactions with the TTAB/LCA/Fe^{3+} system.

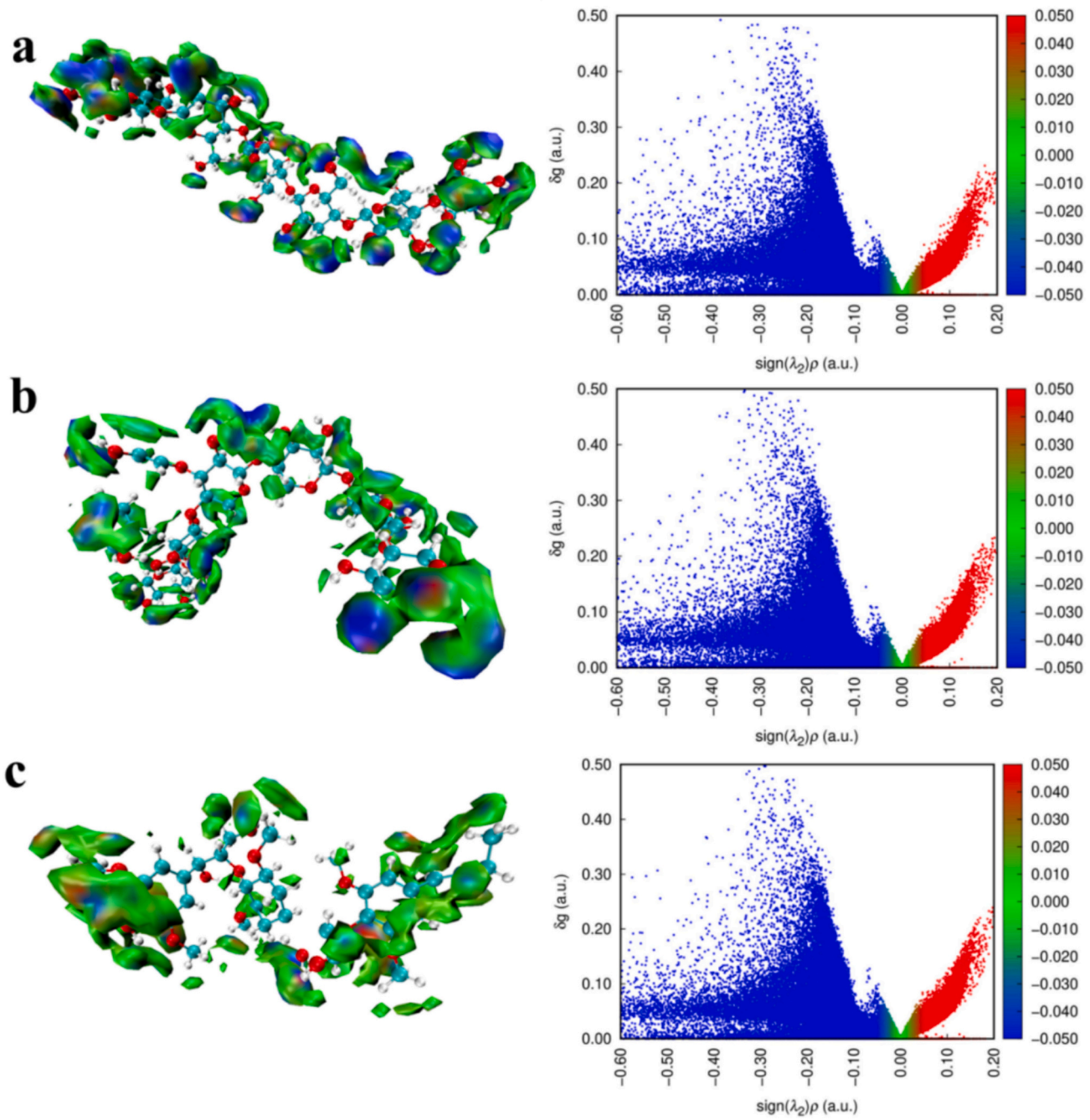


Fig. 8. The independent gradient model (IGM) isosurface image and scatter plot of cellulose model (a), hemicellulose model (b), and lignin model (c) during TTAB/LCA/Fe^{3+} pretreatment.

3.6. Mass balance and energy flow

The pretreatment process of rapeseed straw (RS) involved several key states to efficiently convert and extract valuable components from the biomass (Fig. 9). In the initial state, 1000 g raw RS contained 301.6 g glucan, 151.4 g xylan, and 189.0 g lignin. Subsequently, pretreatment with the TTAB/LCA/Fe^{3+} system effectively eliminated 89.2 % of lignin and 77.9 % of xylan, retaining 79.8 % of glucan. In this progress, 37.5 g of furfural and 36.4 g of XOSs produced from 118.0 g of RS xylan, while 240.8 g of glucan, 33.3 g of xylan, and 20.3 g of lignin were remained in the residues. Furthermore, enzymatic hydrolysis was employed to release the remaining glucan and xylan components, resulting in the release of 247.5 g of glucose and 34.2 g of xylose. In this pretreatment process, carbohydrates in the raw material (RS) were primarily upgraded to reducing sugars, furfural, and XOSs.

A comprehensive analysis of the flow of lignocellulosic energy is necessary to fully understand the utilization of carbohydrates [67]. To further testify the potential of TTAB/LCA/Fe^{3+} as an alternative to pretreatment solvents, the energy flow for this study was calculated. In this work, heat values were calculated based on the mass of different substances and higher heating values (HHV). As showcased in Fig. 10, the heat values of components in lignocellulosic biomass, such as glucan (5.24 MJ), xylan (2.66 MJ), and lignin (4.38 MJ) were considered as energy input. The energy output from the pretreatment process included glucose (4.16 MJ), furfural (0.88 MJ), XOSs (0.66 MJ), pentoses (0.78 MJ), lignin (3.9 MJ), and other by-products (0.06 MJ), demonstrating the versatility of the TTAB/LCA/Fe^{3+} system in producing a range of valuable products from lignocellulosic biomass. Moreover, by

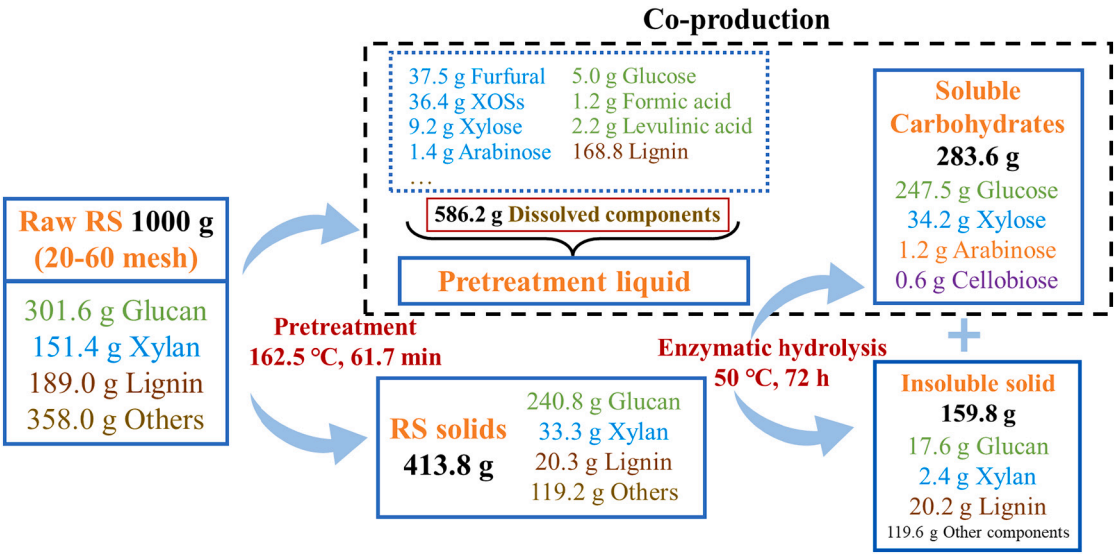


Fig. 9. Mass balance from TTAB/LCA/Fe^{3+} pretreatment to enzymatic hydrolysis of RS.

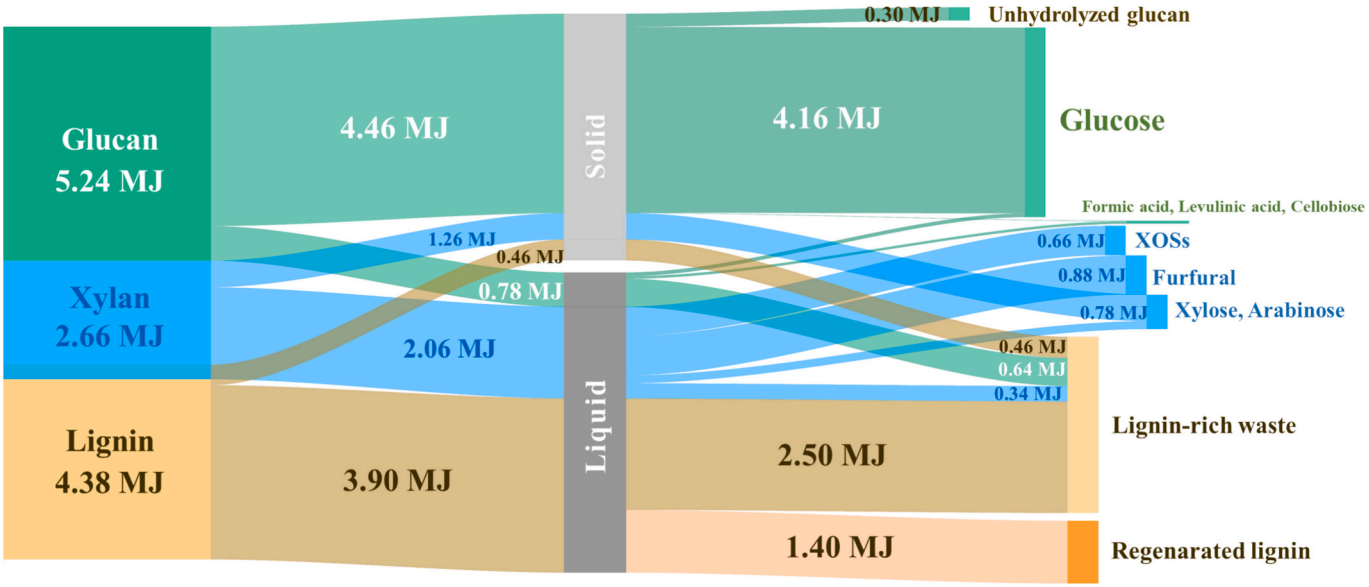


Fig. 10. Energy flow on the co-production of glucose, furfural, and XOSs.

regenerating lignin from the hydrolysate, 36 % of the lignin that was dissolved during pretreatment was obtained. Therefore, the energy recovered from the lignin is about 1.40 MJ. It revealed that the pretreatment process effectually utilized 81.3 % of the energy contained in glucan and 87.2 % of the energy in xylan [67]. The majority of the carbohydrate energy was transformed into value-added forms, indicating the potential for industrial-scale application [67].

In this work, a novel surfactant-based DES system was constructed, which could realize the efficient pretreatment of RS. During the pretreatment process, furfural, XOSs, and reducing sugars were co-produced through the effectual conversion of carbohydrates in biomass. The pretreatment of RS with ternary DES comprising C_nTAB, lactic acid, and $FeCl_3$ demonstrated a promising process for biomass valorization (Fig. 11a). The interaction between Fe^{3+} and lactic acid (LCA) led to the formation of a "Br^--lactic acid-Fe^{3+}" supramolecular complex with the iron ion, while the ternary DES created. In this DES system, LCA provided protons for the efficient hydrolysis of hemicellulose and lignin, facilitating the breakdown of their complex structures. Meanwhile, the surfactant C_nTAB also played a crucial role in the process by enabling delignification and preventing pseudo-lignin recondensation and settling on the biomass surface (Fig. 11b). Fe^{3+} acted as a Lewis acid catalyst, catalyzing the hydrolysis of β-1,4-glycosidic and β-O-4 linkages in hemicellulose and lignin. It underwent coordination reactions with hydroxyl and phenolic groups in lignin and altered its structure and properties making it more vulnerable (Fig. 11b). Although Fe^{3+} might inevitably undergo hydration reaction, transforming to $Fe(OH)_3$ during the reaction, the presence of LCA in the system would greatly inhibit this conversion. These three components in DES acted synergistically to catalyze the depolymerization of lignocellulosic structures, stripping lignin and hemicellulose from RS. Quantum chemistry calculations and molecular dynamics simulations manifested that the dispersion interactions promoted the extraction and dissolution of lignin into the DES system, while hydrogen bonding interactions enhanced the reactivity between the DES and hemicellulose, improving the overall hydrolysis efficiency. Simultaneously, the acidic environment created by Bronsted acid and Lewis acid was conductive to the

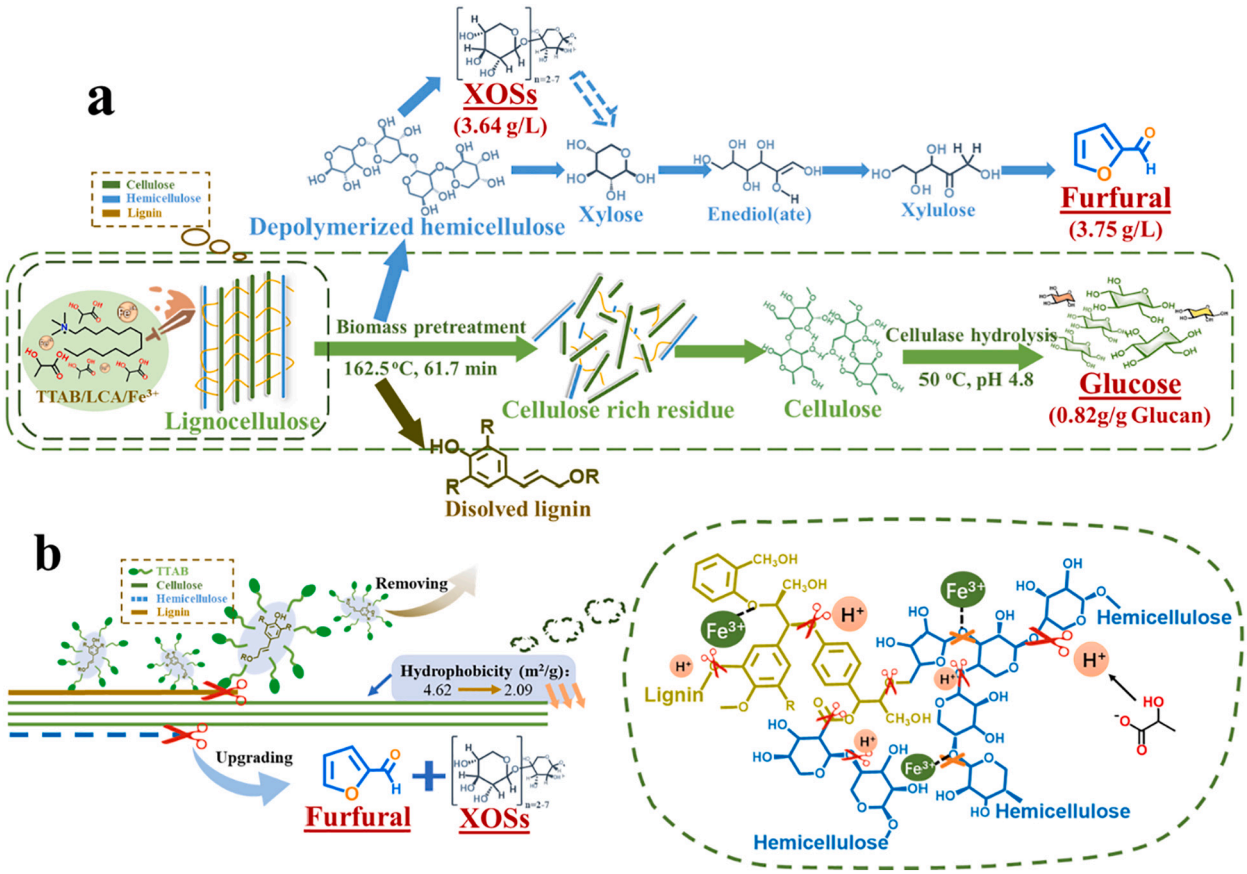


Fig. 11. The mechanism of TTAB/LCA/Fe^{3+} treatment and co-production (glucose, furfural, and XOSs) (a); Detailed mechanism of lignin and xylan elimination during TTAB/LCA/Fe^{3+} treatment (b).

formation of furfural and other high-value byproducts during the pretreatment process. Furfural can be derived into numerous fine chemicals [57]. XOSs are also expensive prebiotic additives widely used in the food industry, promoting human intestinal health [57]. In future studies, more attention will be paid to the development of more cost-effective DESs to establish an economical and effectual pretreatment process for the comprehensive utilization of biomass components [57]. More thorough studies of lignin structure will also be carried out using NMR and confocal Raman microscope to ascertain the effect of TTAB/LCA/Fe^{3+} solvent on lignin structure and to suggest a more thorough pretreatment mechanism in order to obtain a more detailed pretreatment mechanism for directing strategy optimization. To enable further high-value utilization, it was crucial to discuss the methods for separating these two compounds. Furfural separation technologies are primarily divided into distillation, liquid-liquid extraction, and supercritical fluid extraction [68]. Among these, distillation is the most widely used in industrial production due to its mature process and stable product quality. Liquid-liquid extraction is simple, less investment, and low energy consumption. Supercritical fluid extraction has low energy consumption and good separation effects but is hindered by high equipment costs. If extraction methods were to be employed, due to the complex composition of the hydrolysate and the low concentration of furfural, not only would selectivity be poor, but multiple extractions might also be required, increasing production costs and energy consumption. At the same time, the application of supercritical fluid extraction is more difficult due to equipment investment and other reasons. As a result, it seems more feasible to use distillation in this work.

XOSs separation and purification technologies include chromatographic separation, membrane separation, and precipitation methods [69]. In this work, XOSs are present in the pretreatment liquid containing a large amount of lignin, causing significant interference with separation and purification. Generally speaking, membrane separation technology is simple to operate with low energy consumption, strong scalability, and is suitable for industrial production applications. However, due to high equipment and membrane material costs, the initial investment is relatively high; and issues such as membrane fouling may be encountered, requiring regular maintenance. In comparison, chromatographic separation technology offers high purity and strong selectivity, which can effectively achieve the high-purity separation of a single component of oligosaccharides in this work. The typically long operation time can also be accomplished though separation equipment with higher throughput. However, the actual separation and purification technology to be adopted still needs further research based on specific production scales and expected outcomes.

4. Conclusion

In this research, the novel Brønsted and Lewis acid-containing DES TTAB/LCA/Fe^{3+} (1:4:0.0111, mol:mol:mol) was synthesized with TTAB, lactic acid, and $FeCl_3$. Due to the synergistic effect of the three components of DES and the complex hydrogen-bond network, TTAB/LCA/Fe^{3+} showed greatly enhancement of glucose yield and effectually elimination of lignin and xylan at the optimal pretreatment conditions (162.5 °C, 61.7 min). Furthermore, co-production of furfural and XOSs was achieved through the elimination of xylan in RS. In-deep insights of RS pretreatment were given through mass balance and energy flow analysis. The physicochemical properties of DESs were studied, demonstrating its excellent properties of low viscosity, low surface tension, and strong proton donor. A series of characterization and molecular dynamics simulation were used to demonstrate the structural

changes of RS and elucidate the potential pretreatment mechanism at the molecular level, holding significant promising in biomass biorefining.

CRediT authorship contribution statement

Zhengyu Tang: Writing – original draft, Software, Methodology, Investigation, Data curation. **Chaowei Zhang:** Methodology, Investigation. **Junyao Yin:** Methodology, Investigation. **Bo Fan:** Software, Data curation. **Yu-Cai He:** Writing – review & editing, Supervision. **Cuiluan Ma:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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